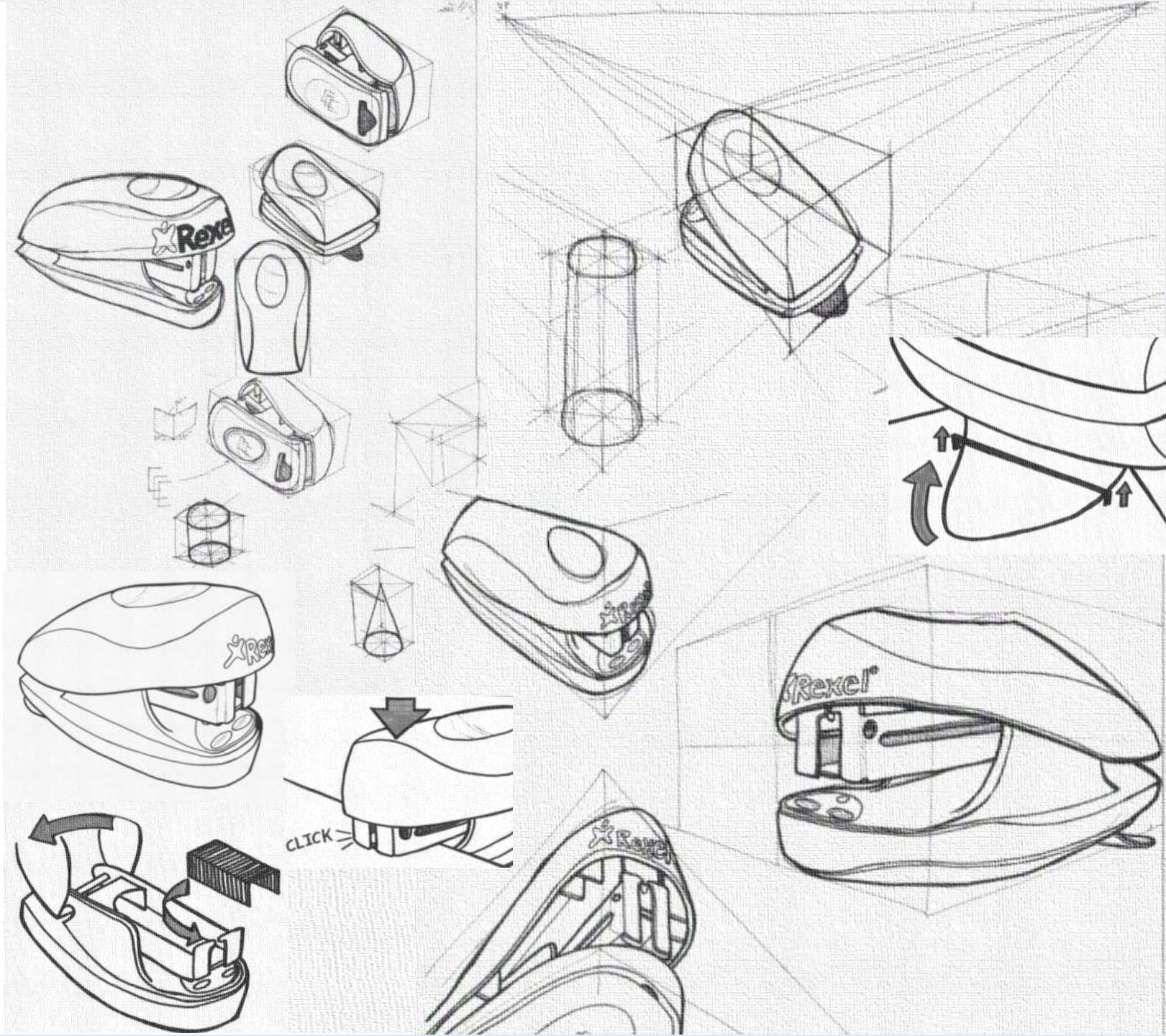
PRODUCT DESIGN
ENGINEERING
PORTFOLIO

Monica Enkhbat

End-to-end product development—from user research, complex 3D CAD to validated physical prototypes and marketing the final product.



1 – Profile & Methodology

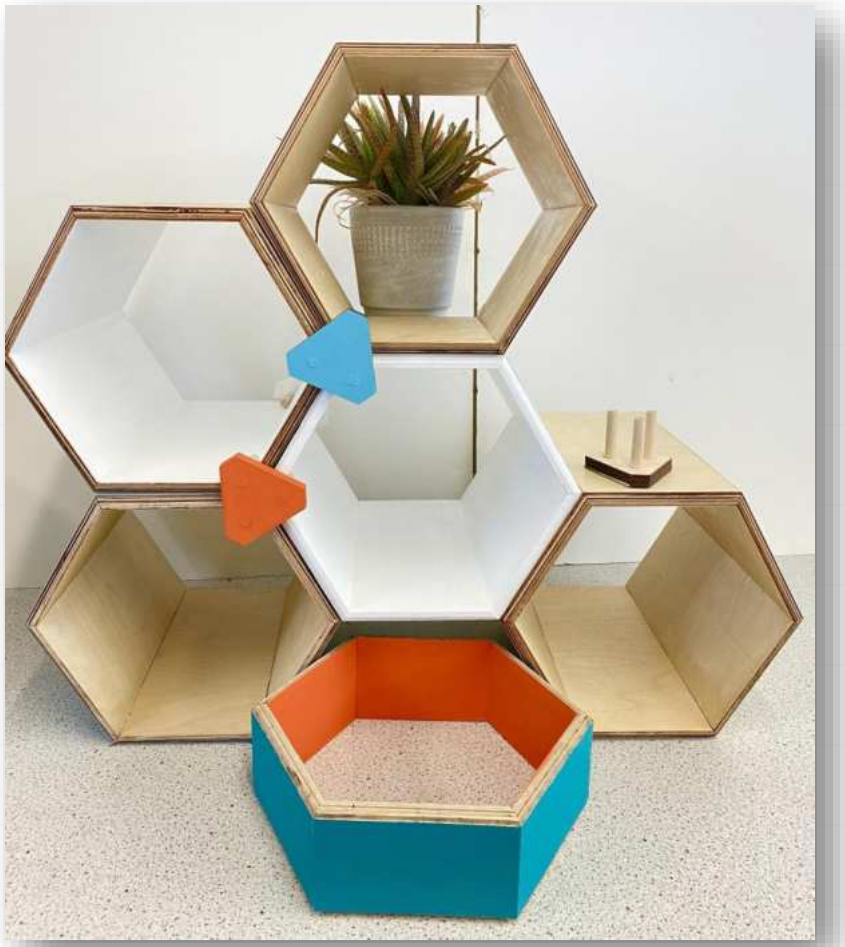


I am a **London-based** Product Design Engineering graduate from Loughborough University specializing in blending visual communication with technical execution.

Engineering & Empathy

- **User-Centred Problem Solving:** Driven by a human-centred approach, focusing on deep user research to solve complex, everyday challenges.
- **Technical & CAD Proficiency:** Skilled in Siemens NX 12.0 modelling, assembly design, and Sustainable Lifecycle Engineering.
- **End-to-End Development:** Experienced across the whole product lifecycle, from initial ideation and storyboarding to physical prototyping and manufacturing considerations.

2 - Foundations & Intention



My interest in design engineering started with a personal challenge: **workspace organization**.

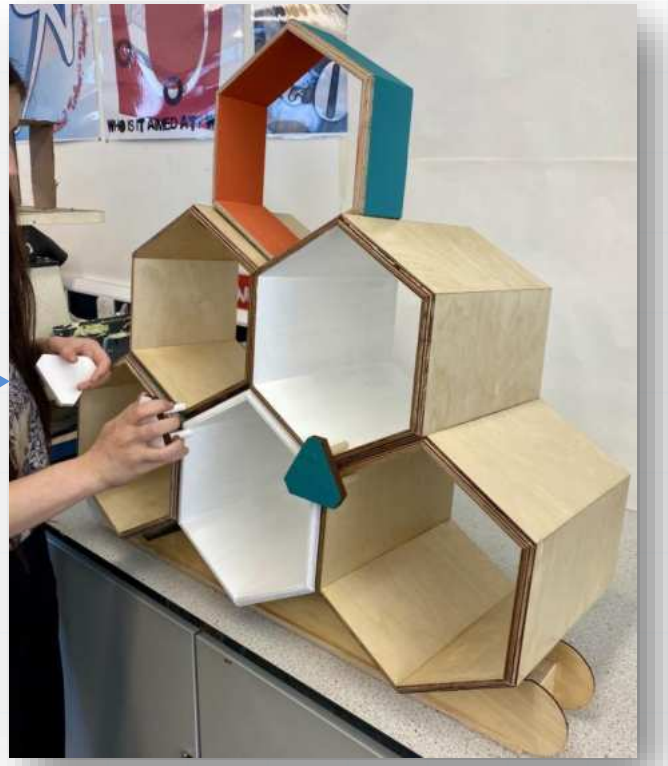
Designing a modular desk system showed me how physical products subtly influence human behaviour. Today, I apply that same human-centred empathy and iterative problem-solving across all engineering disciplines.

Design Intention

I became particularly interested in **designing with intention**, considering user behaviour, motivation, and emotional response, showed how thoughtful design can improve everyday experiences.



[My **client** through the project]



3 – Case Studies

01



02



03





CASE STUDY 01

Medical Laser Tool

Design an ergonomic **medical laser handpiece** with an integrated fume extraction system to capture harmful surgical plumes at the source, complying with British safety standards and improving practitioner comfort during procedures.

Monica Enkhbat

CASE 01 – Ideation & Ergonomics

Form Explorations

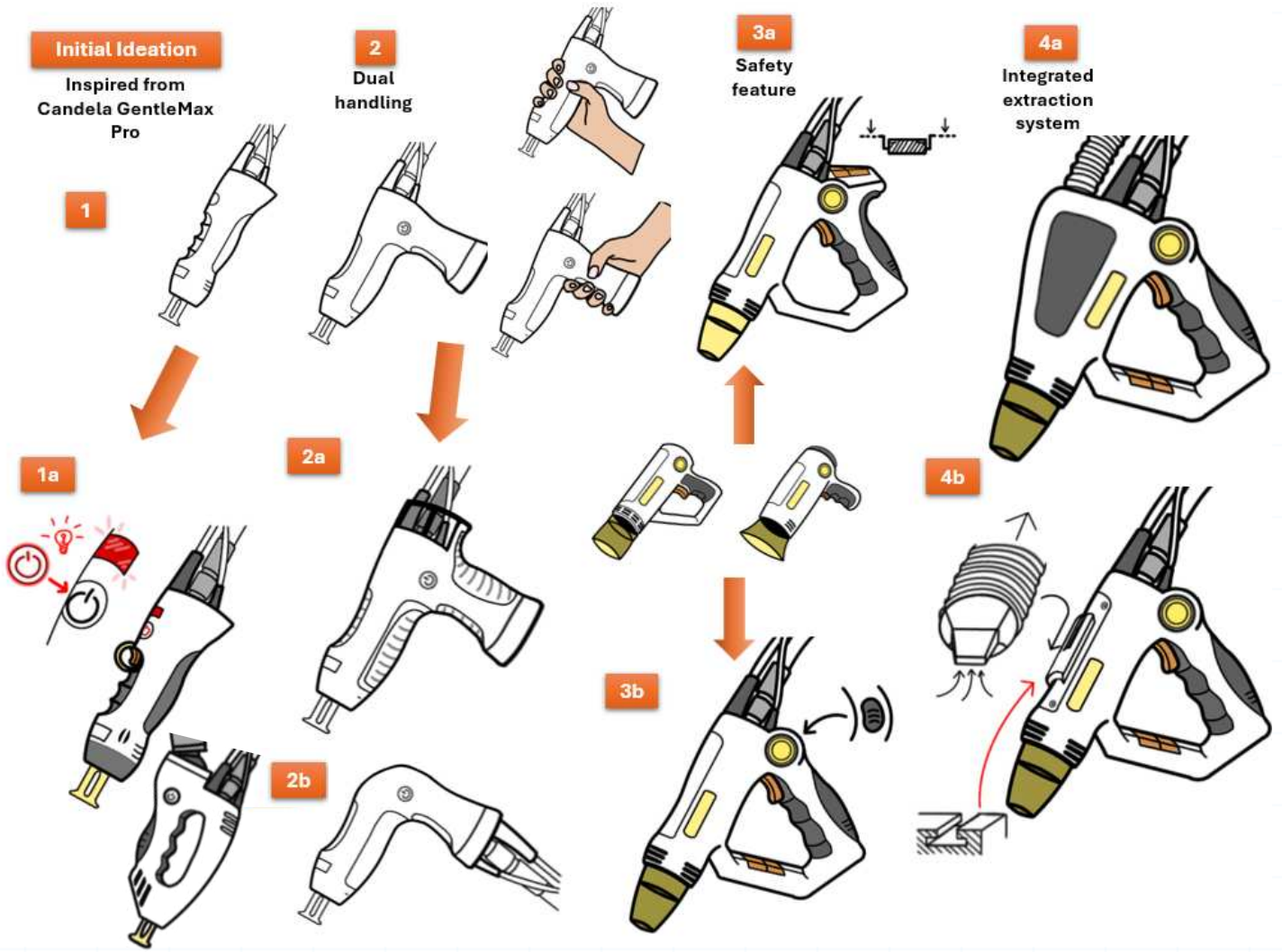
Early handpiece ideation exploring structural geometry, weight balance, and intuitive grip placement. Tested **pen-style**, **dual-grip**, and **pistol-grip** forms to improve comfort and control during use.

Design Elements

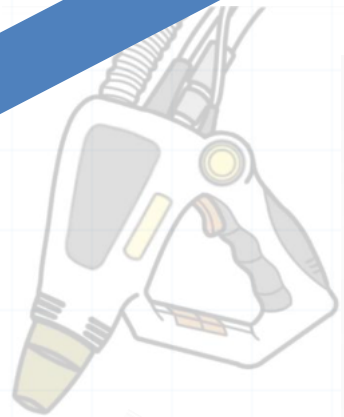
Angled Handle Geometry: Reduce wrist extension strain during long clinical procedures.

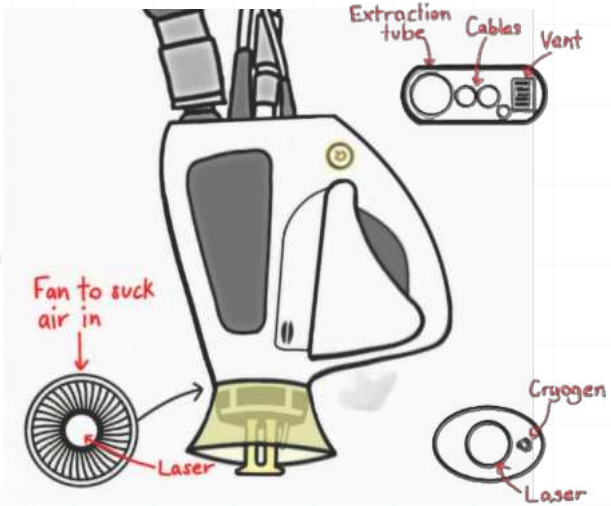
Extraction Placement: Explored different nozzle shapes and tube positions for plume capture.

Candela-Inspired Architecture: Balance grip and ergonomics while housing internal cables.

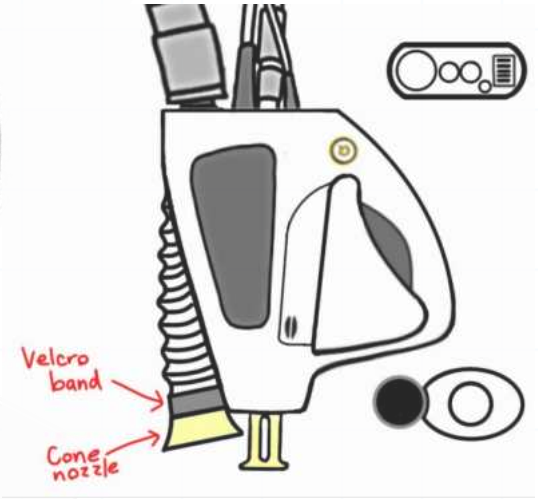


CASE 01 – Concept Iterations

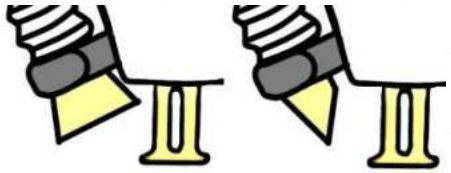


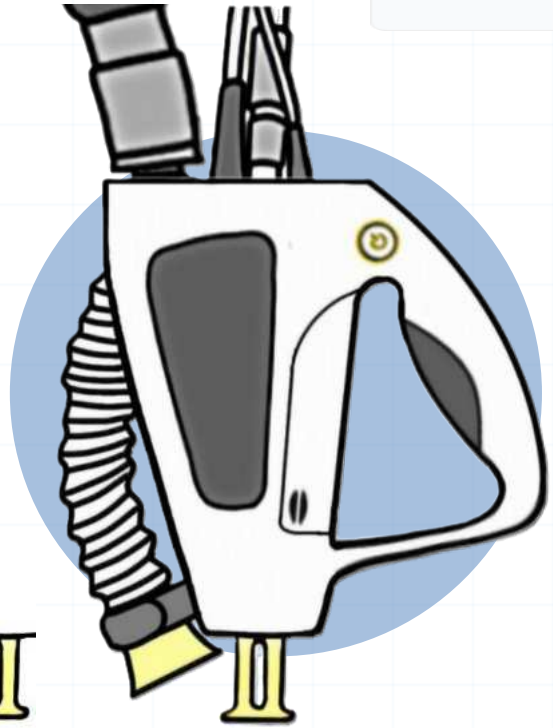












Concept to Final Design

Iterating the handpiece geometry and nozzle assembly to seamlessly combine fume extraction, cable routing, and operator comfort into a refined final form.

Features

- **Optimised Airflow Path:** Refined cone nozzle and extraction tube layout for maximum plume capture.
- **Streamlined Cable Management**
- **Form & Grip Refinement:** Finalised housing contours to support natural hand positions during extended use.

CASE 01 – CAD Architecture

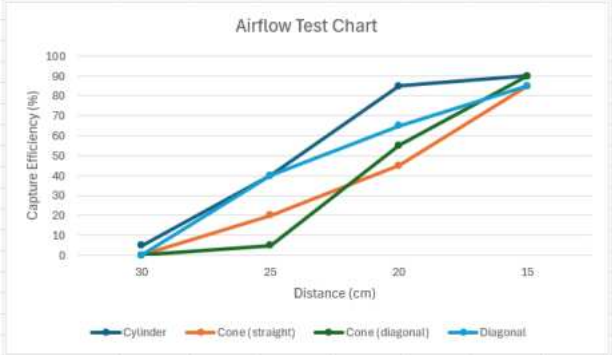
Flow Optimisation

3D modelled in Siemens NX and engineered to capture 85% of procedure fumes. Optimised nozzle geometry maximises airflow velocity and intake efficiency directly at the source.

Video ID	Nozzle Type	Orientation	Distance (cm)	Capture Efficiency (%)	Observations
1	Fail test	Straight	5	99	Setup issue: too close to the source of the smoke.
1	Fail test	Straight	10	95	
1	Fail test	Straight	15	90	
2	Cylinder	Straight	15	90	Smaller diameter, more concentrated suction.
2	Cylinder	Straight	20	85	
2	Cylinder	Straight	25	40	
2	Cylinder	Straight	30	5	
3	Cone	Straight	15	85	Less power in suction due to wider diameter of the nozzle.
3	Cone	Straight	20	45	
3	Cone	Straight	25	20	
3	Cone	Straight	30	0	
4	Cone	Diagonal	15	90	Diagonal: better orientation to suck up rising fumes from above.
4	Cone	Diagonal	20	55	
4	Cone	Diagonal	25	5	
4	Cone	Diagonal	30	0	
5	Diagonal cut	Long side top	15	85	Long side top helps prevent escaped fumes from rising.
5	Diagonal cut	Long side top	20	65	
5	Diagonal cut	Long side top	25	40	
5	Diagonal cut	Long side top	30	0	
6	Diagonal cut	Long side bottom	15	80	More escaped fumes gliding over the nozzle, nozzle too open at the top side.
6	Diagonal cut	Long side bottom	20	60	
6	Diagonal cut	Long side bottom	25	20	
6	Diagonal cut	Long side bottom	30	0	

	30	25	20	15
Cylinder	5	40	85	90
Cone (straight)	0	20	45	85
Cone (diagonal)	0	5	55	90
Diagonal	0	40	65	85

	70-99%
	50-69%
	1-49%



Tables & Chart: Airflow testing to determine optimal nozzle shape and distance.



CASE 01 – NX Human Modelling



Demographic Scaling: Fits 5th percentile female to 95th percentile male anthropometrics.

Grip Geometry: Contoured handle profile reduces wrist strain during prolonged use.

Reach & Clearance: Optimised trigger and hose angles for physical operation.







CASE 01 – Prototype Validation

Virtual

Evaluated ergonomics and use in a **clinical environment**.

Physical

3D-printed split-shell prototypes to verify structural balance and **real-world handling**.






25WSC501
Laser+
2026



CASE STUDY 02

HydroHero

Develop an engaging, user-centred **hydration bottle** that resolves daily usability frustrations: eliminating leaks, easing operation, and providing passive level tracking to encourage healthy drinking habits.

CASE 02 – Client Diary

Research & TRIZ Analysis

Tracked daily water volume reduction over a 5-day client diary alongside consensual daily interviews with both child and parent to identify real-world usage habits.

Pain Points

User Frustration (Child): Leakage and awkward lid mechanisms discouraged drinking during use.

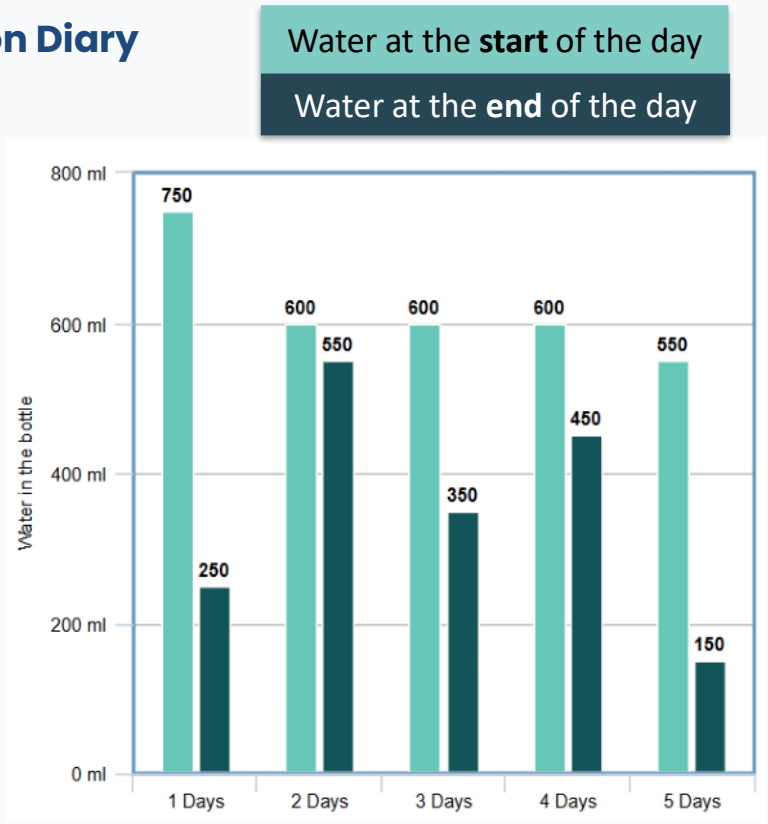
Parent Frustration: Unfinished water leads to spills, mess, and unnecessary waste.

TRIZ Contradiction

*"Improve **Ease of Operation** without contradicting **Device Complexity**"*

- Increase **reminder visibility** without adding **electronics** or **user complexity**.

5-Day Hydration Diary



1. Tue/Thu

Perception vs. Reality: Difference between the child's perceived intake from the interview and actual intake from the data.

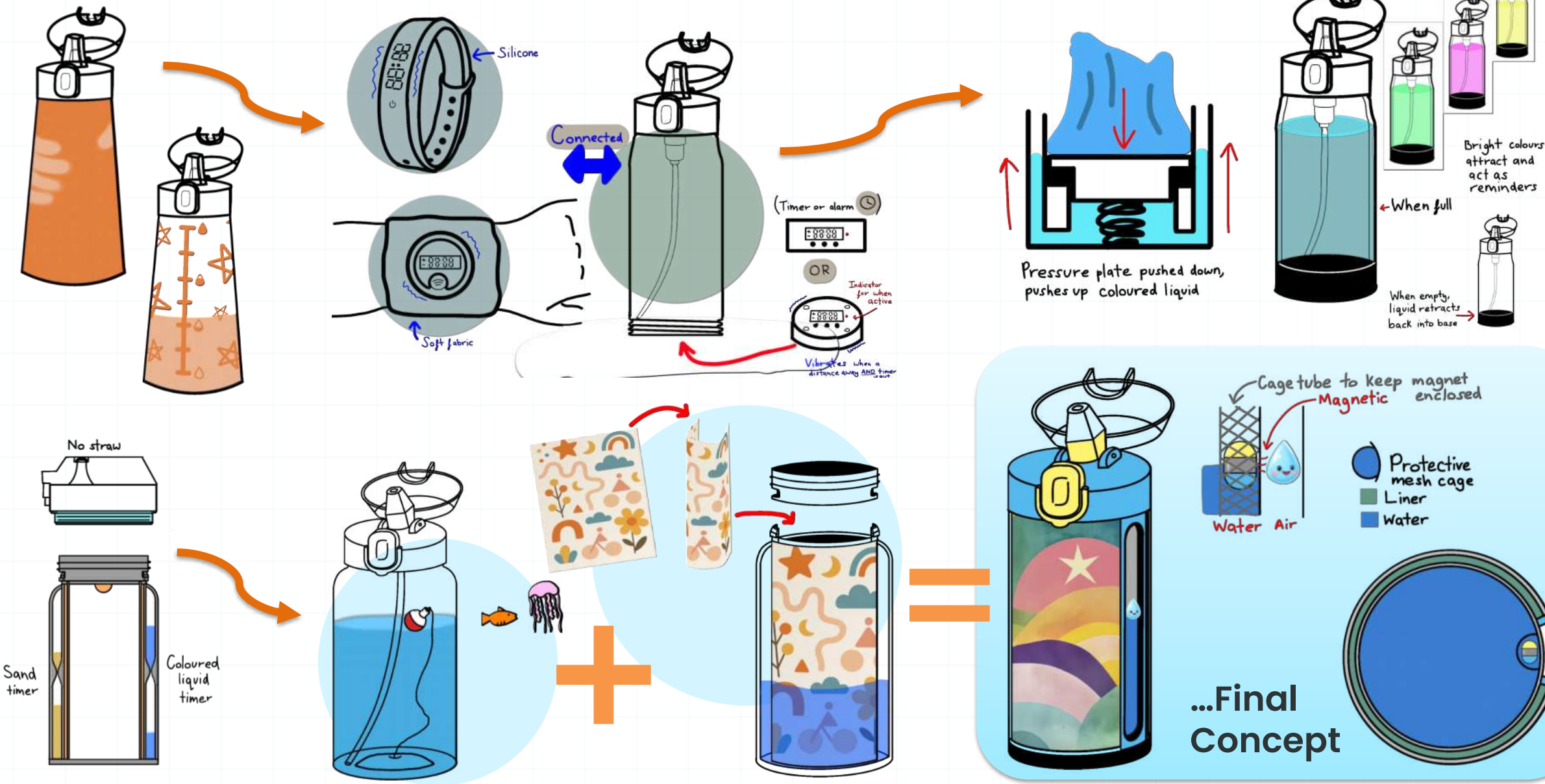
2. Tue

Mechanical Loss: Observed that a stiff screw-lid acted as a physical barrier and demotivator to the child.

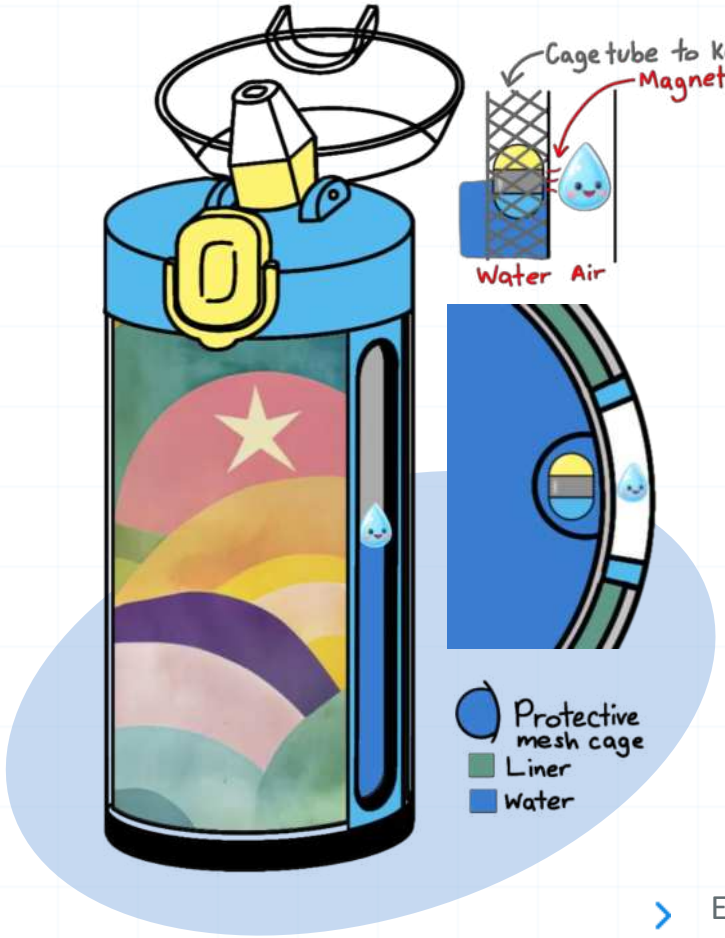
3. Wed

Sensory Rejection: Identified that metallic odour build-up from the metal bottle caused immediate rejection, leaving the bottle over half full.

CASE 02 – Concepts Journey



CASE 02 – CAD & Mechanism

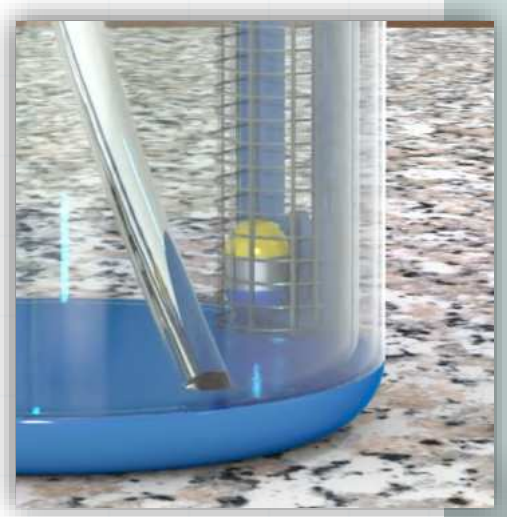


Magnetic Float Tracking

- Modelled an **internal buoyant magnetic bobber** that pairs through the bottle wall to drive the external level indicator passively **without** electronics.
- **Protective mesh enclosure** to keep the magnetic float secure, preventing choking hazards or loss during use and washing.

Liner Housing

- Engineered a dual-wall casing to allow custom interchangeable graphic liners.





CASE 02 – Product Features

HYDROHERO

Spill-proof silicone straw

Magnetic buoyant float delivers **real-time water level feedback**

Engaging interchangeable liners

Push-button flip lid and **lock**

Hygienic **Odour-Free** Bottle: wide-neck, plastic body eliminates metallic smells and **eases cleaning**









sagentia innovation

CASE STUDY 03

Pawfect Portions

(Sagentia Innovation Challenge)

Design an **automated pet feeding solution** to maintain optimum health in household dogs by encouraging portion control, reducing food waste, and applying Design for Manufacture and Assembly (DFMA) principles.

Monica Enkhbat

CASE 03 - Aesthetic Development

The Problem

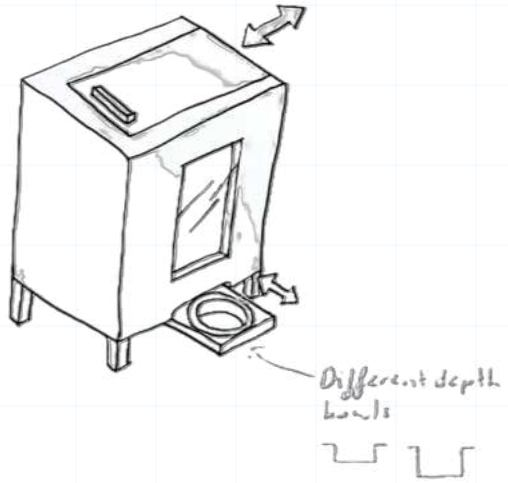
Overfeeding and inaccurate portioning contribute directly to canine **obesity** and household pet food **waste**.

Conventional gravity feeders **lack controlled dispensing**, leading to unmonitored intake.

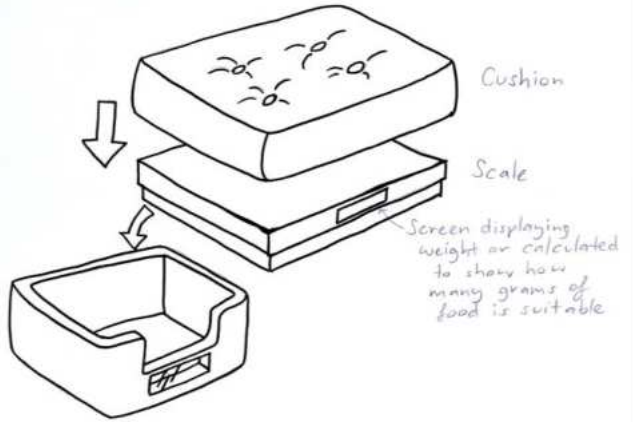
Concept Selection

Generated **75** initial concept sketches and evaluated each on a 1-5 scale across the **PDS** and **specification**, systematically filtering the concepts.

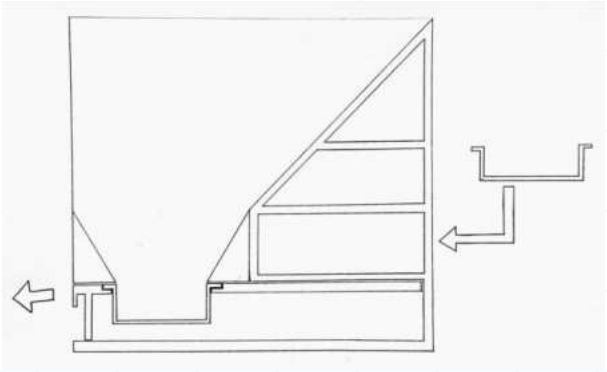
1) Gravity Hopper Exterior



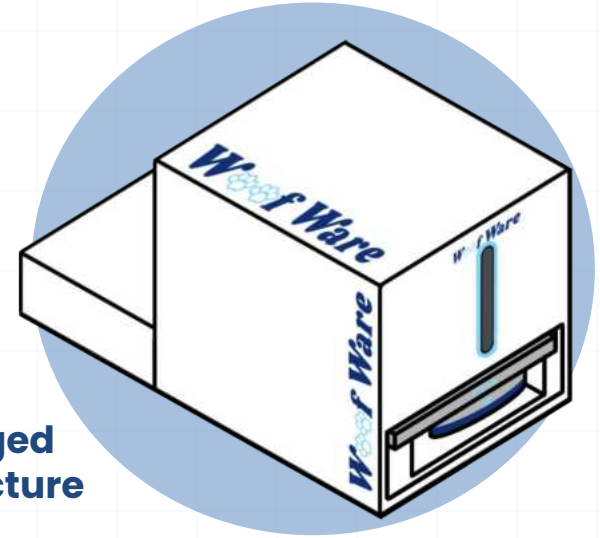
2) Smart Weight-Tracking Bed



3) Gravity Hopper Sloped Interior

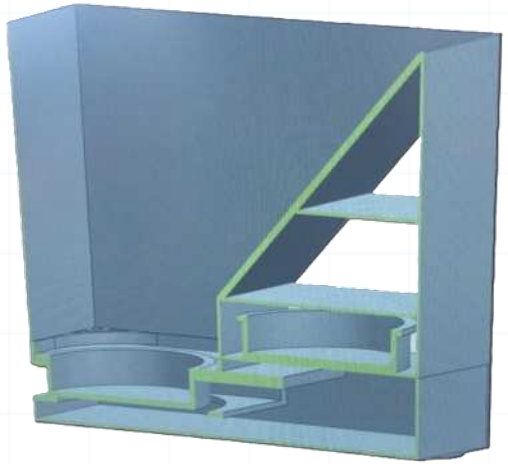


Final Converged Architecture



CASE 03 – CAD Prototypes



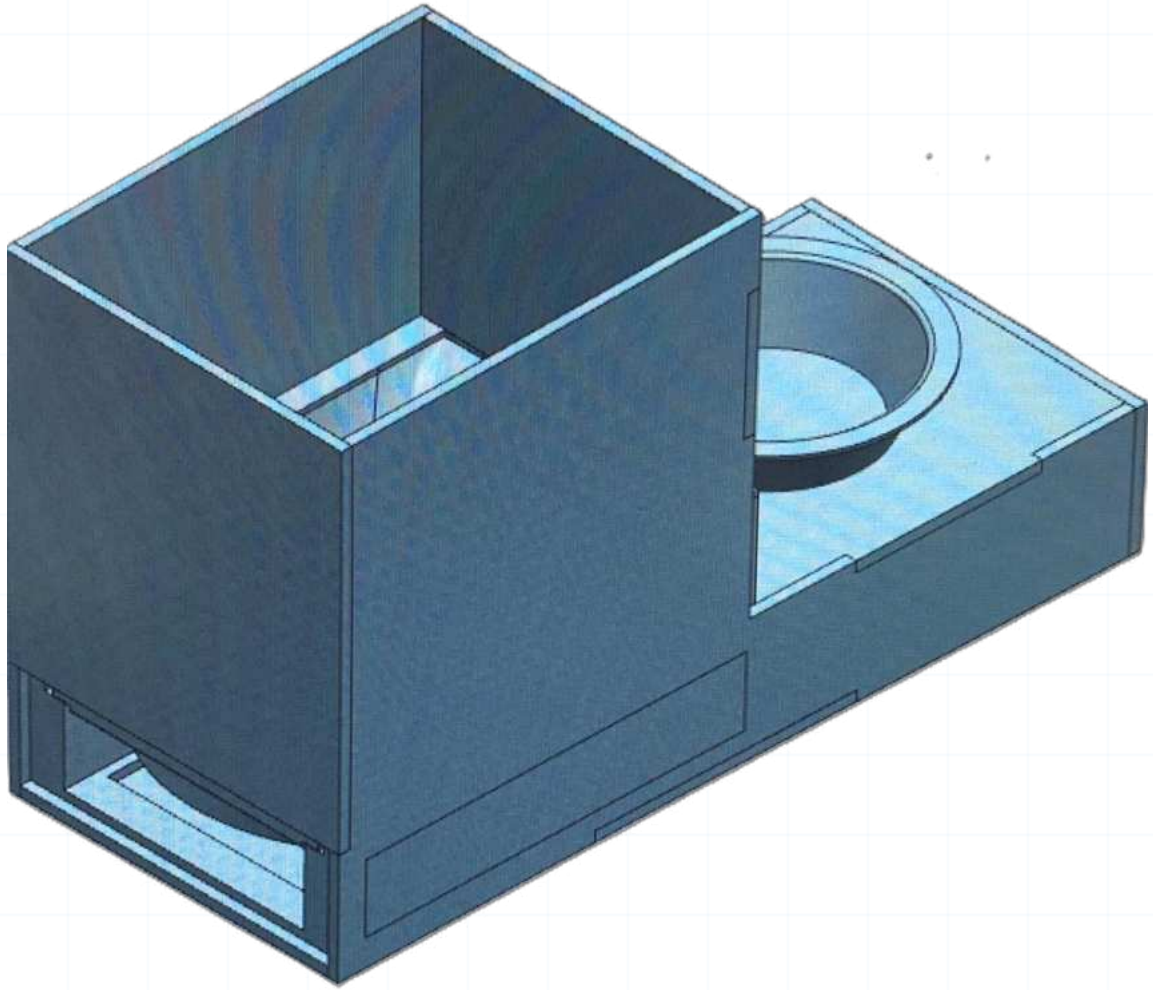


Modelled the hopper, bowls, and drawer mechanism in **CAD** to test how kibble moves through the internal slopes into the sliding tray.

Final CAD Model

DFMA Jointing: Sized panels for flat-sheet manufacture.

Zero-Power: Eliminating electrical failure points.

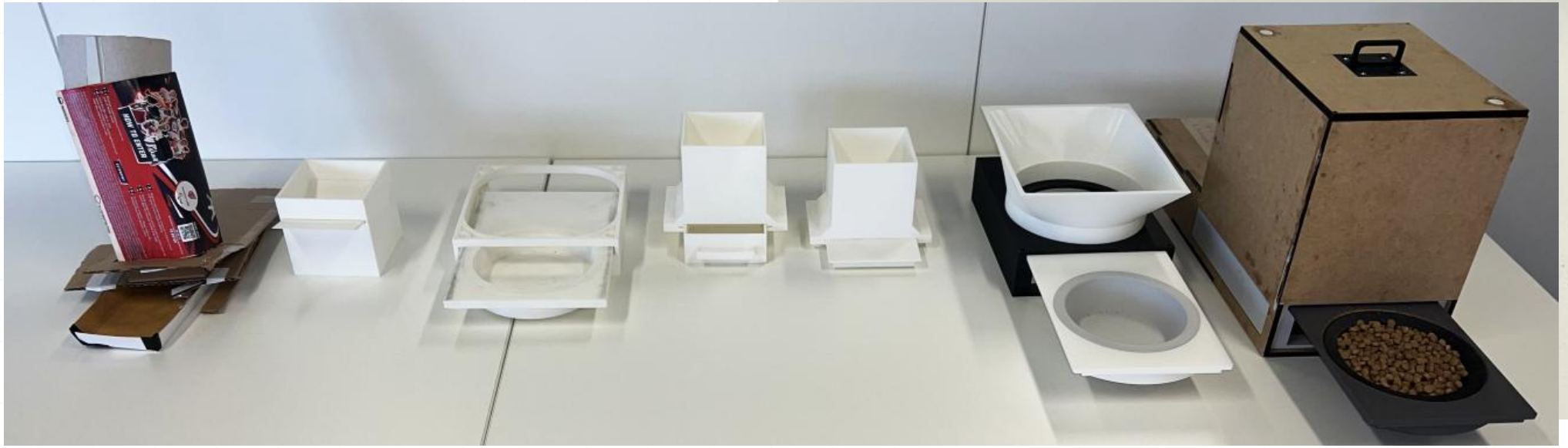


CASE 03 - Prototype Development

Iterations

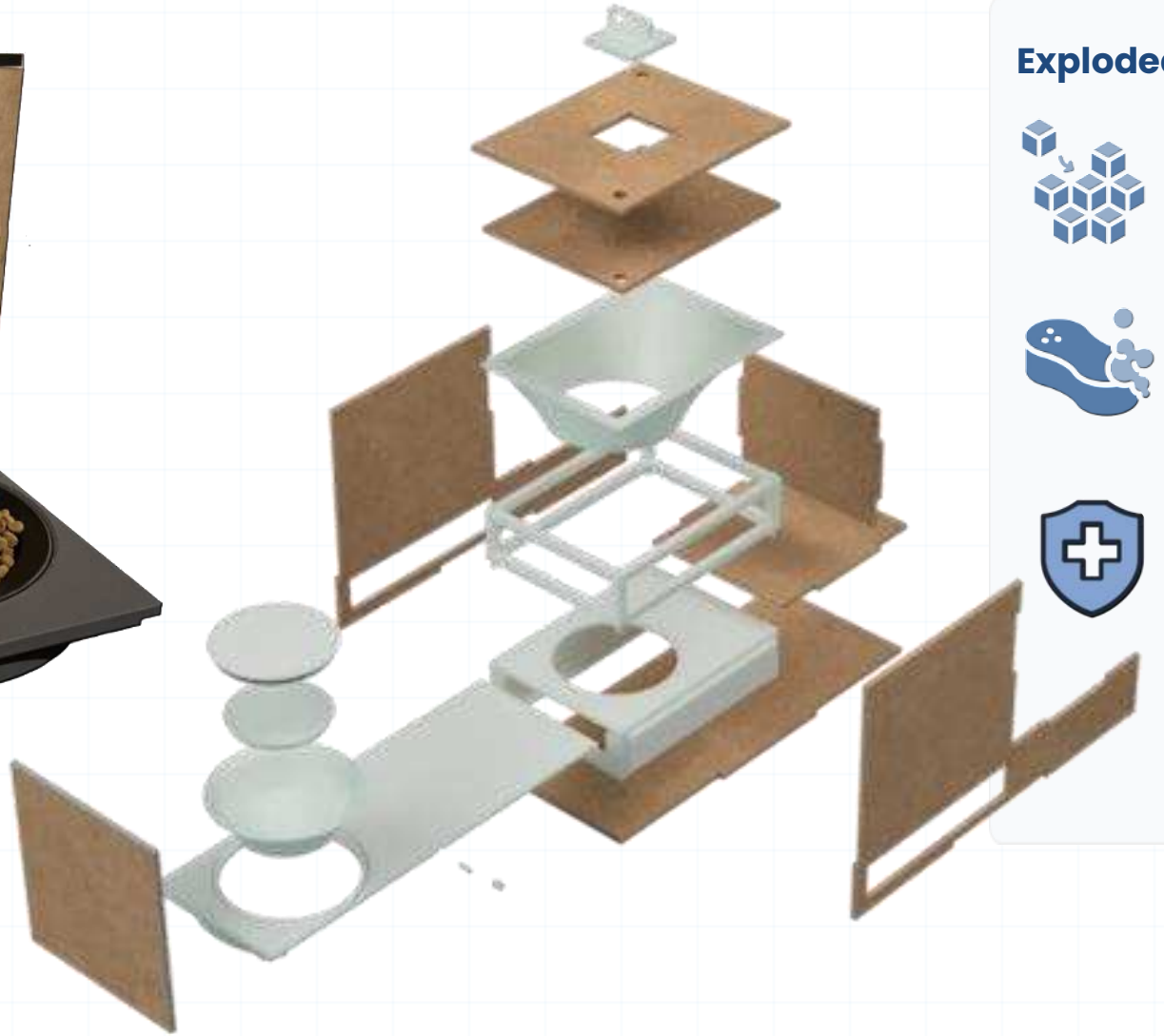
Tested early **cardboard and 3D printed models** to refine the feeder's shape, size, and mechanism. By trying out different 3D printed bowl depths and insert shapes, we optimised portion control for various dog sizes to create the final working prototype.





CASE 03 - Prototype Disassembly





Exploded Assembly & Architecture

Modular Assembly: Laser-cut panels designed for easy component replacement.

Detachable Tray & Bowl: Removable for quick daily washing.

Hygienic Material: Food-safe plastic interior paired with sustainable outer panels.

CASE 03 – Visual Language

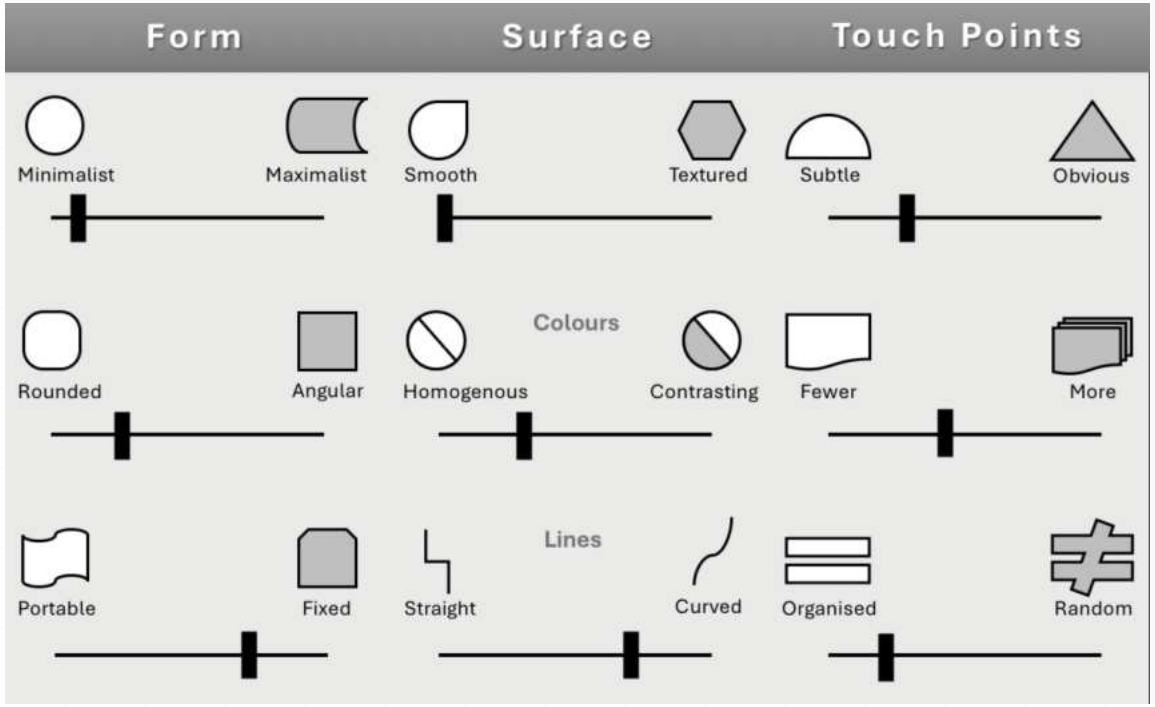
Mood Boarding

Defined a soft architectural visual language with matte materials to integrate into **home environments**.



Style Sliders

Balanced "playful" vs "minimal" attributes to align with **WoofWare brand expectations**.



CASE 03 – Product Ecosystem

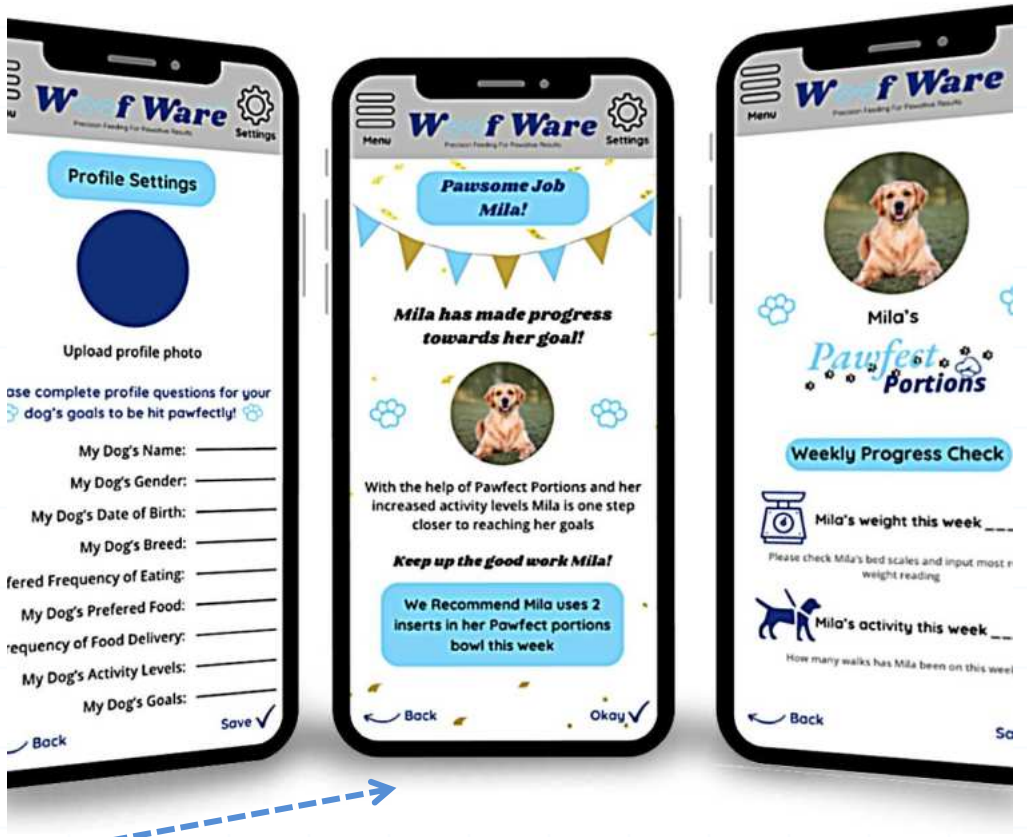
Product Eco

Developed an intuitive app ecosystem to track feeding data and suggest bowl inserts.



- Tailored pet health through an intuitive phone app by taking the data from the smart dog bed.

WoofWare



- The app would inform the user of how many bowl inserts to use.

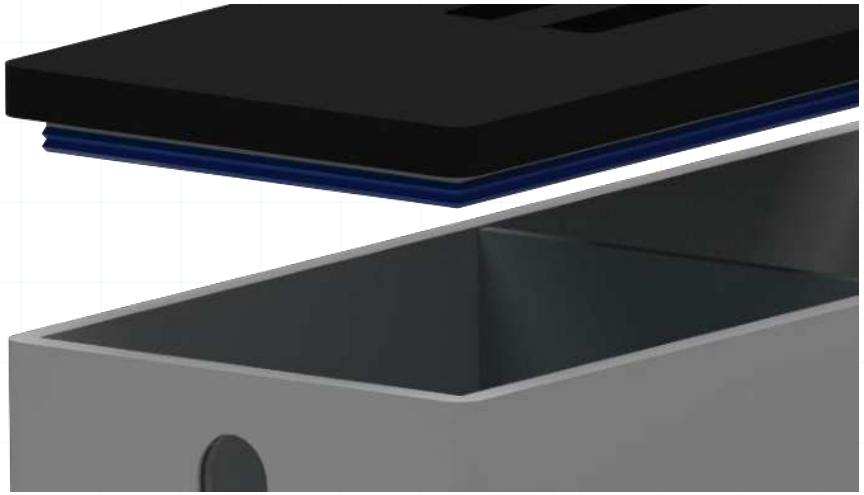
CASE 03 – Product Features

Modular sizing: Scaled feeder sizes and **interchangeable bowl** inserts for tailored portions

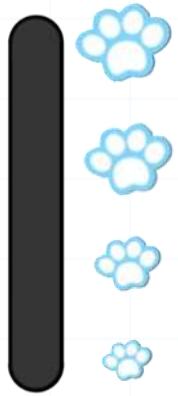
Airtight **silicone seal**

Visual indicator: Clear window for easy **food level** monitoring

Smooth drawer dispensing mess-free portion





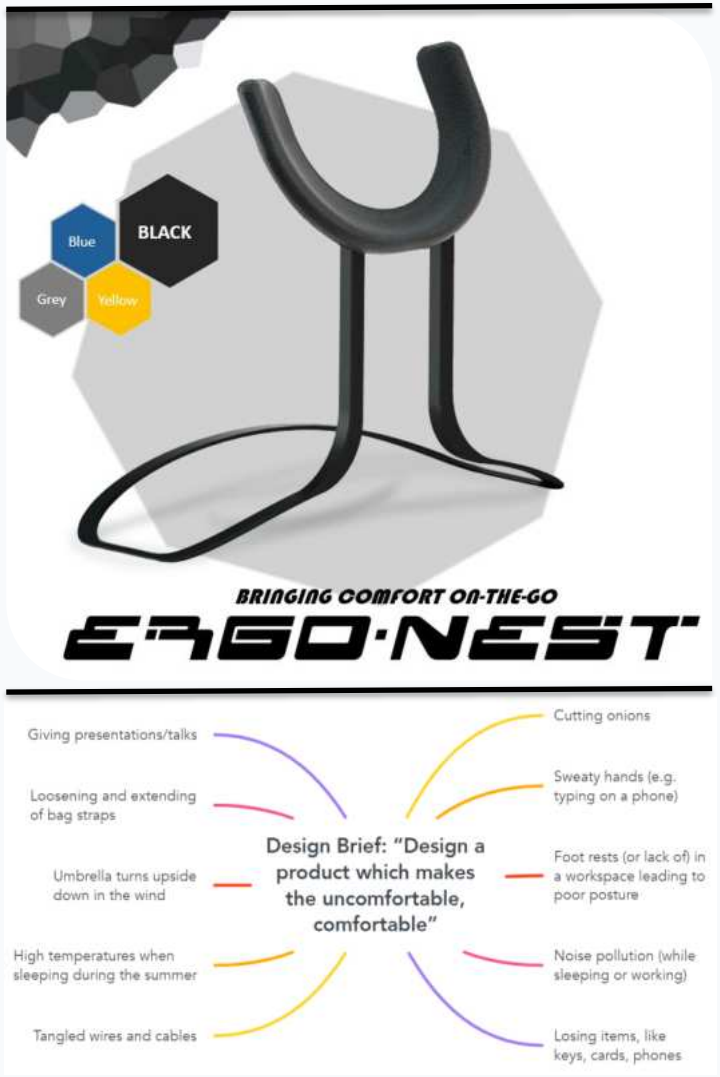


- Food level indicator





4 – Other Projects



Fabuletta





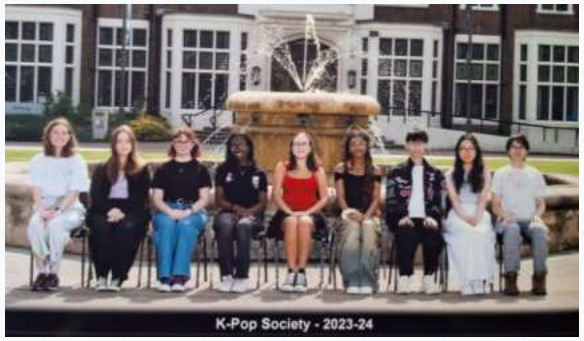


5 – More Branding & Design

Versatile design experience spanning apparel **merchandise** procurement, **local business** visual identity, and visual **communication**.

Merchandise

Society merch from concept to sales.





Branding

Local business leaflets, menus, social media.







Posters

Visual narrative for exhibitions.





Let's Connect

Available for Design Opportunities

Email

monicaenkhbat@gmail.com

LinkedIn

linkedin.com/in/monica-enkhbat